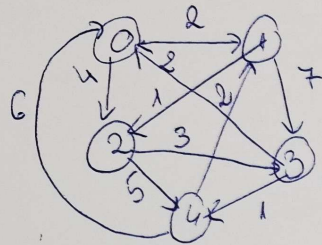


Dijkstra algorithm - graph with no negative costs and existing walk

S=0 T=4

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The minimum cost walk from 0 to 4 is build backwards from previous

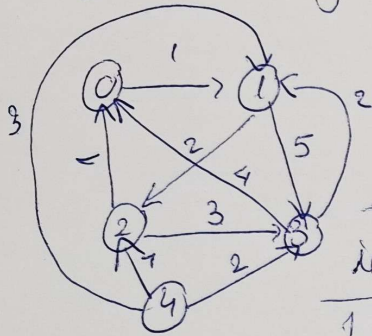
$T=4$, $prev[4]=3$,
 $prev[3]=2$, $prev[2]=1$,
 $prev[1]=0$
 $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$
 $cost = dist[4] = 7$

	x	y	distance	priority queue	previous
init			0	$\leftarrow [0,0] \leftarrow$	
iter 1	0			$\leftarrow \leftarrow$	1
iter 1.1		1	0 1	$\leftarrow [2,1] \leftarrow$	0
iter 1.2		2	0 1 2	$\leftarrow [(2,1) (4,2)] \leftarrow$	1 2 0 1 2
iter 2	1			$\leftarrow [(4,2)] \leftarrow$	
iter 2.1		2	0 1 2	$\leftarrow [(3,2) (4,2)] \leftarrow$	1 2 0 1 2
iter 2.2		3	0 1 2 3	$\leftarrow [(3,2) (4,2) (9,3)] \leftarrow$	1 2 3 0 1 1 1
iter 3	2			$\leftarrow [(4,2) (9,3)] \leftarrow$	
iter 3.1		3	0 1 2 3	$\leftarrow [(4,2) (6,3) (9,3)] \leftarrow$	1 2 3 0 1 1 2
iter 3.2		4	0 1 2 3 4	$\leftarrow [(4,2) (6,3) (8,4) (9,3)] \leftarrow$	1 2 3 4 0 1 1 2 1 2
iter 4	2	-	(no update)	$\leftarrow [(6,3) (8,4) (9,3)] \leftarrow$	
iter 5	3			$\leftarrow [(8,4) (9,3)] \leftarrow$	1 2 3 4 0 1 1 2 1 3
iter 5.1	3	4	0 1 2 3 4	$\leftarrow [(7,4) (8,4) (9,3)] \leftarrow$	
iter 6	4			X=4	

Dijkstra algorithm - graph with no negative costs and no walk

$S=0$ $T=4$

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group 913



	x	y	distance	priority queue	previous
init			$\begin{matrix} 0 \\ 0 \end{matrix}$	$\leftarrow [0,0] \leftarrow$	
iter 1	0		$\begin{matrix} 0 \\ 0 \end{matrix}$	$\leftarrow \leftarrow$	$\begin{matrix} 1 \\ 0 \end{matrix}$
iter 1.1		1	$\begin{matrix} 0 & 1 \\ 0 & 1 \end{matrix}$	$\leftarrow [0,0] \leftarrow$	$\begin{matrix} 1 \\ 0 \end{matrix}$
iter 2	1		$\begin{matrix} 0 & 1 \\ 0 & 1 \end{matrix}$	$\leftarrow \leftarrow$	$\begin{matrix} 1 & 2 \\ 0 & 1 \end{matrix}$
iter 2.1		2	$\begin{matrix} 0 & 1 & 2 \\ 0 & 1 & 2 \end{matrix}$	$\leftarrow [3,2] \leftarrow$	$\begin{matrix} 1 & 2 & 3 \\ 0 & 1 & 1 \end{matrix}$
iter 2.2		3	$\begin{matrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 3 & 6 \end{matrix}$	$\leftarrow [3,2] [6,3] \leftarrow$	$\begin{matrix} 0 & 2 & 3 \\ 0 & 1 & 1 & 1 \end{matrix}$
iter 3	2			$\leftarrow [6,3] \leftarrow$	$\begin{matrix} 0 & 2 & 3 \\ 0 & 1 & 1 & 1 \end{matrix}$
iter 3.1		3	no update		$\begin{matrix} 0 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 & 2 \end{matrix}$
iter 4	3			$\leftarrow \leftarrow$	
iter 4.1		1	no update		

The priority queue is empty $\Rightarrow \text{dist}[4] = \text{infinity} \Rightarrow$ the vertex 4 is not reachable